

CAAM/STAT31470: Homework #7

Purpose: The goal of these questions is to provide you practice for calculations and reinforce concepts from class. Unless specifically told in the question, you should always show your working and justify your answers. Marks will be awarded for your reasoning in addition to the correctness of your answers. Also note that these questions will be similar to the questions on the Midterm and Final.

Content: For the final two weeks we look at applying complex analysis to finding integrals that might appear in applications. In particular, we see how computing residues and using the Cauchy residue formula can be used to compute integrals without any obvious real analytic means.

Homework is due: **Friday May 23, 11:59pm.**

Problem 7.1 (4 points)

Suppose $f(z)$ is analytic at z_0 and $g(z) = f(z)/(z - z_0)$. Then explain why:

- (a) If $f(z_0) \neq 0$ then z_0 is a simple pole of g with residue $f(z_0)$.
- (b) If $f(z_0) = 0$ then z_0 is a removable singularity of g .

Problem 7.2 (4 points)

Show that $e^{1/z} = -1$ whenever $z = -i/((2n+1)\pi)$ for any n in $\mathbb{Z}$. For what values of $c \in \mathbb{C}$ can you find infinitely many solutions to the equation $e^{1/z} = c$ near $z = 0$? Hence, is there a well-defined limit of $e^{1/z}$ as z tends to 0?

Problem 7.3 (4 points)

Write the function

$$f(z) = \frac{8a^3 z^2}{(z^2 + a^2)^3}$$

for $a > 0$ as

$$f(z) = \frac{\phi(z)}{(z - ai)^n}$$

where $\phi(z)$ is analytic and non-zero at $z = ai$.

- (a) Why does $\phi(z)$ have a Taylor series at $z = ai$?
- (b) Find the principal part of $f(z)$ at $z = ai$.

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Problem 7.4 (9 points)

Let C_N be the positively oriented square whose edges lie on $x = \pm(N + 1/2)\pi$ and $y = \pm(N + 1/2)\pi$ for an integer N which is large.

(a) Using the Cauchy residue theorem show

$$\int_{C_N} \frac{dz}{z^2 \sin z} = 2\pi i \left[\frac{1}{6} + 2 \sum_{n=1}^N \frac{(-1)^n}{n^2 \pi^2} \right]$$

(b) Show

$$\lim_{N \rightarrow \infty} \int_{C_N} \frac{dz}{z^2 \sin z} = 0$$

(c) Hence, conclude that

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$$

Problem 7.5 (15 points)

The goal of this question is to compute the integral

(1)
$$\int_0^{\infty} \frac{x^{-a}}{x+1} dx$$

for a real number $0 < a < 1$. This is an improper integral because we must take limits both as x goes to infinity and to zero.

(a) Why can we not integrate directly around the contour in Figure 1?

(b) Apply the Cauchy residue theorem to the branch

(2)
$$f_1(z) = \frac{z^{-a}}{z+1}, \quad (|z| > 0, -\frac{\pi}{2} < \arg z < \frac{3\pi}{2})$$

by integrating around the contour in Figure 2(A) to find

(3)
$$\int_{\rho}^R \frac{r^{-a}}{r+1} dr + \int_{\Gamma_R + L + \Gamma_{\rho}} f_1(z) dz = 2\pi i \operatorname{Res}_{z=-1} f_1(z)$$

(c) Apply the Cauchy-Goursat theorem to the branch

(4)
$$f_2(z) = \frac{z^{-a}}{z+1}, \quad (|z| > 0, \frac{\pi}{2} < \arg z < \frac{5\pi}{2})$$

by integrating around the closed contour in Figure 2(B) to find a similar expression to equation (3).

(d) Explain why we can replace f_1 and f_2 in the expressions obtained above by the branch

(5)
$$f(z) = \frac{z^{-a}}{z+1}, \quad (|z| > 0, 0 < \arg z < 2\pi)$$

and use this to show

$$\int_{\rho}^R \frac{r^{-a}}{r+1} dr + \int_{C_R} f(z) dz - \int_{\rho}^R \frac{r^{-a} e^{-i2a\pi}}{r+1} dr + \int_{C_{\rho}} f(z) dz = 2\pi i \operatorname{Res}_{z=-1} f_1(z)$$

(e) By evaluating the residue and taking the limit as $R \rightarrow \infty$ and $\rho \rightarrow 0$, determine the value of the integral.

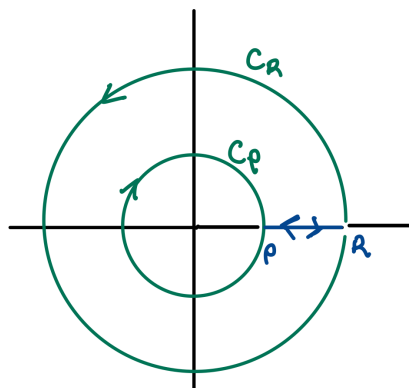


Figure 1: C_R is the positively oriented circle of radius R . C_ρ is the negatively oriented circle of radius ρ and we assume $\rho < 1 < R$.

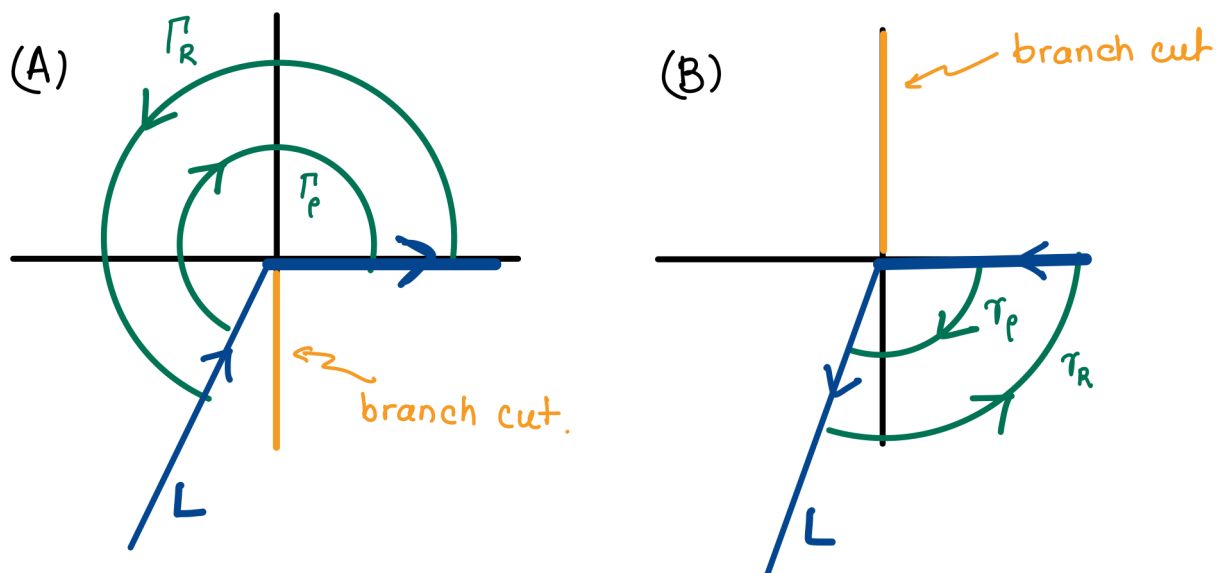


Figure 2: We divide the contour in figure 1 into two pieces along the line L . Then $C_R = \Gamma_R + \gamma_R$ and $C_\rho = \Gamma_\rho + \gamma_\rho$. In the integral expressions, I use L to mean the small segment of the line joining the circle segments.